

SOLUTION REQUIREMENTS

Visualization Tool for Electric Vehicle Charge and Range Analysis

1. Functional Requirements

The system should:

- Display electric vehicle specifications such as range, efficiency, battery capacity, and price.
- Display charging station information.
- Allow users to compare electric vehicle brands.
- Provide interactive filters for better analysis.
- Display dashboards and stories using Tableau.
- Allow visualization of EV performance using charts and maps.
- Support data imported from MySQL database.

2. Non-Functional Requirements

- Dashboard should be easy to use.
- Reports should load quickly.
- Data should be accurate and reliable.
- Dashboard should support future dataset expansion.
- Users should easily understand the visualizations.
- The solution should provide consistent performance.

3. Hardware Requirements

Processor: Intel Core i3 or above

RAM: 4 GB or above

Storage: 20 GB Free Space

Internet Connection: Required

Display: 1366×768 Resolution or above

4. Software Requirements

Operating System: Windows 10 / Windows 11

Database: MySQL Workbench

Visualization Tool: Tableau Public

Spreadsheet Tool: Microsoft Excel

Version Control: GitHub

Documentation: Microsoft Word

Conclusion

The above functional and non-functional requirements define the expected behavior of the Visualization Tool for Electric Vehicle Charge and Range Analysis. These requirements ensure that the dashboard is reliable, interactive, and capable of supporting efficient analysis of electric vehicle data.